Week 5 Benchmark Models

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This brief summarizes the Week 5 benchmark models, their performance, and the four metrics used to evaluate both point forecasts and predictive distributions.

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1 Benchmark Models

Autoregressive with Three Lags (RF)
Rolling Average 3-Month (RF)

Both models are estimated on the seven horizon logged fatalities.

1.1 Week 5 benchmark scripts

- `week_5_1_setup_data.R`: Builds lag features and forecast splits.
- `week_5_2_train_rf_lag123.R`: Trains the Lag123 benchmark using `log_lag1, log_lag2, log_lag3`.
- `week_5_3_train_rf_roll3.R`: Trains the Roll3 benchmark using `log_roll3`.
- `week_5_4_add_crps.R`: Computes deterministic CRPS from the point forecasts.
- `week_5_5_probabilistic_crps.R`: Computes probabilistic CRPS, calibration, and sharpness.
- `week_5_6_compare_models.R`: Produces the final model-comparison tables and RMSE/MSE plots.
- `week_5_7_compare_metrics.R`: Produces the final cross-metric comparison outputs and combined figure.

1.2 Evaluation design

- Horizons: 1 to 7 months ahead.
- Target: `log_fatalities_ahead_hm`.
- Cross validation design: time-aware folds (10 folds, 12-month validation windows, minimum 120 months initial train history).
- Benchmark algorithm: random forest through `caret + ranger` with 1000 trees.

2 Benchmark Performance

2.1 RMSE

Horizon	Lag123 RMSE	Roll3 RMSE	Delta (Roll3 - Lag123)
1	0.6316	0.7044	+0.0728
2	0.6554	0.7397	+0.0843
3	0.6711	0.7505	+0.0794
4	0.6848	0.7594	+0.0745
5	0.6973	0.7784	+0.0811
6	0.7069	0.7834	+0.0764
7	0.7185	0.7925	+0.0739

2.2 MSE

Horizon	Lag123 MSE	Roll3 MSE	Delta (Roll3 - Lag123)
1	0.3989	0.4962	+0.0973
2	0.4296	0.5472	+0.1176
3	0.4504	0.5633	+0.1129
4	0.4690	0.5767	+0.1076
5	0.4862	0.6058	+0.1196
6	0.4998	0.6137	+0.1139
7	0.5163	0.6280	+0.1117

2.3 Deterministic CRPS

Horizon	Lag123 CRPS (det)	Roll3 CRPS (det)	Delta (Roll3 - Lag123)
1	0.2604	0.2923	+0.0319
2	0.2740	0.3071	+0.0330
3	0.2830	0.3123	+0.0293
4	0.2917	0.3181	+0.0264
5	0.2977	0.3264	+0.0287
6	0.3011	0.3286	+0.0275
7	0.3062	0.3341	+0.0279

2.4 Probabilistic CRPS

Horizon	Lag123 CRPS (prob)	Roll3 CRPS (prob)	Delta (Roll3 - Lag123)
1	0.2757	0.3209	+0.0452
2	0.2890	0.3353	+0.0463
3	0.2979	0.3411	+0.0432
4	0.3056	0.3467	+0.0412
5	0.3119	0.3542	+0.0423
6	0.3155	0.3569	+0.0415
7	0.3185	0.3602	+0.0417

3 Metric Definitions and Interpretation

3.1 Mean Squared Error (MSE)

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2$$

MSE is the average squared forecast error.

- Lower is better.

- MSE penalizes large misses heavily because errors are squared.

3.2 Root Mean Squared Error (RMSE)

$$\text{RMSE} = \sqrt{\text{MSE}}$$

RMSE returns the score to the same units as the modeled target.

- Lower is better.
- RMSE is the point-forecast comparison metric.

3.3 Deterministic CRPS

$$\text{CRPS}_{\text{det}} = \frac{1}{n} \sum_{i=1}^{n} |\hat{y}_i - y_i|$$

For a predictive distribution, CRPS reduces to absolute error.

- Lower is better.
- Deterministic CRPS scores the point forecast using the same proper-scoring-rule framework as the probabilistic version.

3.4 Probabilistic CRPS

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} \left(F(z) - \mathbb{1}\{y \leq z\}\right)^2 dz$$

- Lower is better.
- Probabilistic CRPS rewards forecasts that place probability mass near the realized outcome.
- It jointly reflects calibration and sharpness.

4 Benchmark Figure

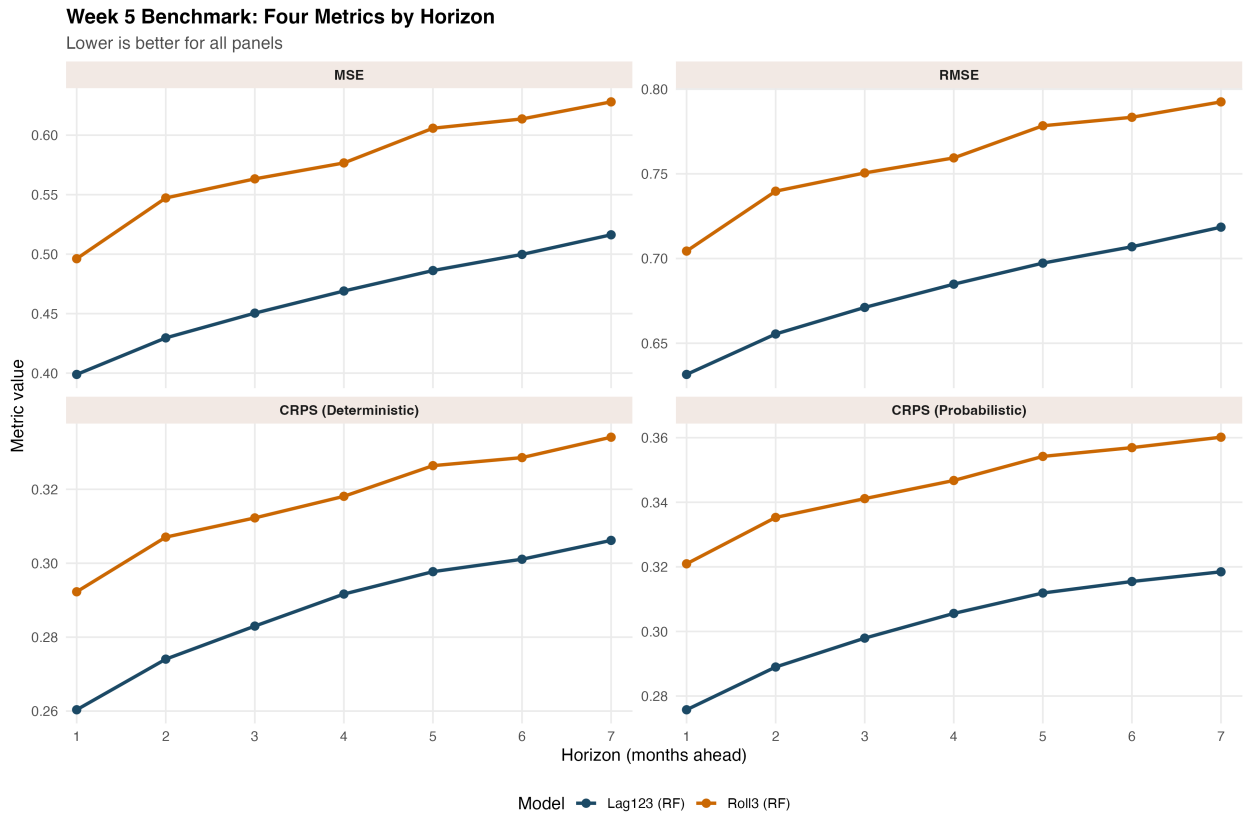

Figure 1: Week 5 benchmark comparison across MSE, RMSE, deterministic CRPS, and probabilistic CRPS. Lower is better in every panel.